

SYDNEY TECHNICAL HIGH SCHOOL



Year 11

PRELIMINARY HIGHER SCHOOL CERTIFICATE

ASSESSMENT TASK 3

SEPTEMBER 2016

Mathematics Extension 1

General Instructions

- Working time – 90 minutes
- Write using black or blue pen
- Approved calculators may be used
- In questions 6 to 11, show relevant mathematical reasoning and/or calculations
- Start each question in section 2 on a new page
- Full marks may not be awarded for careless or badly arranged work

Total marks – 71

Section 1 – 5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section.

Section 2 – 66 marks

Attempt Questions 6 – 11

Allow about 82 minutes for this section.

Name: _____

Teacher: _____

Section 1**5 marks**

Allow about 8 minutes for this section. Use the multiple choice answer sheet in your answer booklet for questions 1 – 5. Do not remove the multiple choice answer sheet from your answer booklet.

1. Which of the following is an expression for:

$$\frac{d}{dx} (x\sqrt{x^2 + 2})?$$

A) $\frac{x}{\sqrt{x^2+2}}$

B) $1 + \frac{x}{\sqrt{x^2+2}}$

C) $\frac{2x^2+2}{\sqrt{x^2+2}}$

D) $\frac{2x^2+x+4}{\sqrt{x^2+2}}$

2. A (-7, 10) and B (1, -2) are two points. What is the value of k such that the point $P(k, 1)$ divides the interval AB internally in the ratio of 3 : 1?

A) -5

B) -3

C) -1

D) 5

3. A curve has parametric equations $x = at$ and $y = \frac{1}{2} (a + at^2)$. What is the Cartesian equation of the curve?

A) $x^2 = 4ay$

B) $x^2 = 2a$

C) $x^2 = a(2y - a)$

D) $x^2 = 2y - a$

4. The graph of the even polynomial passes through the point (1, 2). What is the remainder when $P(x)$ is divided by $(x + 1)$?

- A) -2
- B) -1
- C) 1
- D) 2

5. Which of the following is an expression for:

$$\frac{\sin 4x}{\sin x}$$

- A) $\frac{1}{2} \cos 2x \cos x$
- B) $\cos 2x \cos x$
- C) $2 \cos 2x \cos x$
- D) $4 \cos 2x \cos x$

Section 2 **66 marks**

Attempt questions 6 – 11.

Allow about 82 minutes for this section.

Answer each question in your answer booklet. Start each question on a new page.

Question 6

11 Marks

- a) Make x the subject in the formula

2

$$y = \frac{x - 3}{2 - x}$$

- b) Consider the polynomial $P(x) = x^3 + ax + b$ where a and b are real numbers.
 $(x - 2)$ is a factor of $P(x)$ and when $P(x)$ is divided by $(x + 1)$ the remainder is 6.

- (i) Show that $2a + b = -8$ and $b - a = 7$

2

- (ii) Find the values of a and b .

1

- c) A parabola has equation

$$y^2 - 2y - 8x + 17 = 0$$

- (i) Find the coordinates of its vertex.

1

- (ii) Sketch the parabola showing its x intercept

1

- (iii) On your sketch, display the focus and directrix.

2

- d) The equation $x^3 + 2x^2 + 3x + 1 = 0$ has roots α, β and γ . Find the value of
 $\alpha^2 + \beta^2 + \gamma^2$

2

Question 7 (Start a new page)**11 Marks**

- a) Find the values of a and b if:

2

$$x^2 + 10x - 3 = ax(x + 1) + b(x - 1)^2$$

- b) Given the polynomial $P(x) = -x^3 - ax$ is odd and has $(x + 2)$ as a factor:

Find a

1

- (i) Sketch the graph of $y = P(x)$ showing the intercepts on the x axis.

1

- (ii) Solve the inequality $P(x) \leq 0$

1

- c) (i) Use the substitution $t = \tan \frac{x}{2}$ to show that the equation $3 \sin x - \cos x = 1$ is equivalent to the equation $t = \frac{1}{3}$

2

- (ii) Solve the equation $3 \sin x - \cos x = 1$ over the domain $0^\circ \leq \theta \leq 360^\circ$ giving the answers correct to the nearest minute.

2

- d) Solve the inequality $\frac{1}{x} - \frac{1}{x-2} > 0$

2

Question 8 (Start a new page)**11 Marks**

- a) Solve the equation $\sqrt{2} \cos(x + 60^\circ) - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$

3

- b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$ with parameter values $p = 2$ and $q = -1$. Find the angle made correct to the nearest degree between the tangents at these points.

2

- c) If $\sec x - \tan x = k$ for some real number k , show that $\sec x + \tan x = \frac{1}{k}$

2

- d) Show that $3x - 4y + 10 = 0$ is a tangent to the circle $x^2 + y^2 = 4$

2

- e) (i) Show that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$.

1

- (ii) Hence or otherwise evaluate $2 \sin 45^\circ \cos 15^\circ$ and leave your answer in exact form.

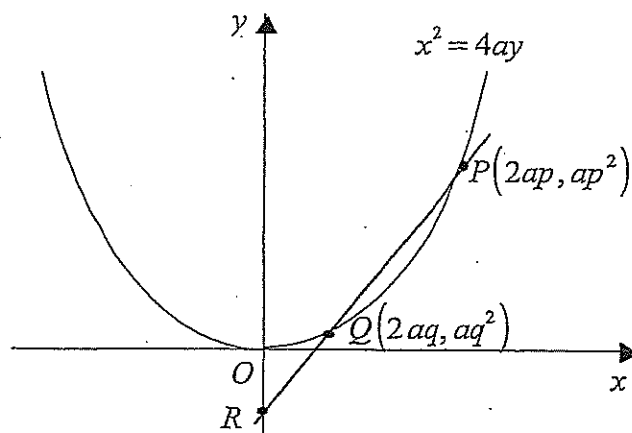
1

Question 9 (Start a new page)

11 Marks

- a) (i) Sketch $y = |x - 2|$ 1
- (ii) Hence or otherwise solve $|x - 2| = 2 - x$ 1

b)



$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

- (i) Show that the chord PQ has equation $(p + q)x - 2y = 2apq$. 2
- (ii) If P and Q move on the parabola such that $pq = 1$, where $p \neq 0$ and $q \neq 0$, show that the chord PQ (produced) always passes through a fixed pt. R on the y axis. 1
- c) Find all values of k which will make the expression: 2
- $(k + 1)x^2 - 2(k - 1)x + (k - 5)$ a perfect square
- d) The polynomial equation $x^3 + bx^2 + cx + d = 0$ has roots α, α^2 and α^3 for some real number $\alpha \neq 0$
- (i) Find in terms of c and d the value of $\frac{1}{\alpha} + \frac{1}{\alpha^2} + \frac{1}{\alpha^3}$. 2
- (ii) Show that $b^3d - c^3 = 0$ 2

a) For what value(s) of x is the function $y = \frac{x+1}{x-1}$ not differentiable and give a brief reason why not. 1

b) For $f(x) = \frac{x^2}{x^2-4}$

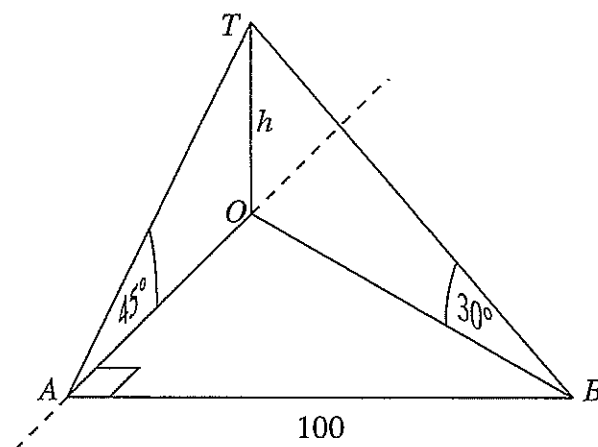
(i) Find $\lim_{x \rightarrow \infty} \frac{x^2}{x^2-4}$ 1

(ii) Show that it is an even function 1

(iii) State the domain 1

(iv) Without using calculus, sketch the curve showing all important features. 2

c)



A surveyor stands at a point A , which is due south of a tower OT of height h m. The angle of elevation of the top of the tower from A is 45° . The surveyor then walks 100 m due east to a point B , from where she measures the angle of elevation of the top of the tower to be 30° .

(i) Express the length of OB in terms of h . 1

(ii) Show that $h = 50\sqrt{2}$ 2

(iii) Calculate the bearing of B from the base of the tower, correct to the nearest degree. 2

Question 11 (Start a new page)

11 Marks

- a) Given $(x - 3)$ is a factor of the polynomial $f(x) = 2x^3 - 7x^2 - 7x + 30$, find all solutions for $f(x) = 0$. 2
- b) Find the coordinates of the point on the curve $y = x^2 + 3x - 1$ where the tangent is parallel to the line $y = 5x + 6$. 2
- c) $A(2ap, ap^2)$ and $B(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.
- (i) Find the coordinates of the midpoint M of the chord joining $A(2ap, ap^2)$ and $B(2aq, aq^2)$ in simplified and factorised form. 1
- (ii) Given the equation of the chord is $y = \frac{1}{2}(p+q)x - apq$ and it passes through the focus of the parabola, show that $pq = -1$. 1
- (iii) Hence show by substitution that M lies on the parabola $x^2 = 2ay - 2a^2$. 3
- (iv) Find the vertex and focus of the parabola given in part (iii). 2

END OF TEST

Student Name: _____

Teacher Name: _____

2016 Year 11 Ext. 1 Final Prelim. Solutions

Section 1

1. C 2. C 3. C 4. D 5. D

Section 2

6a) $y = \frac{x-3}{2-x}$

$2y - xy = x - 3$

$2y + 3 = x + xy$

$2y + 3 = x(1+y)$

$x = \frac{2y+3}{y+1}$

b) i) $P(2) = 8 + 2a + b = 0$

$P(-1) = -1 - a + b = 6$

$\therefore 2a + b = -8$

$b - a = 7$

ii) $3a = -15$

$a = -5 \therefore b = 2$

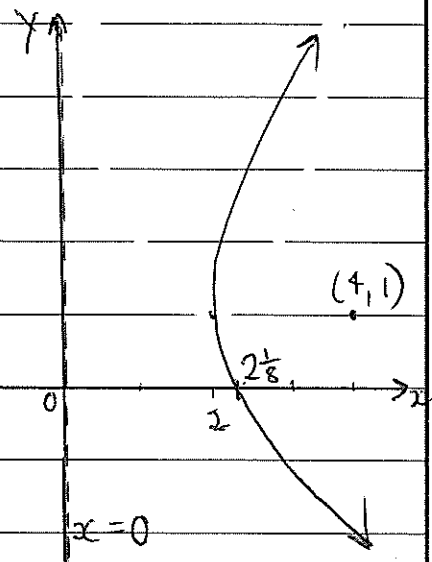
c) i) $y^2 - 2y = 8x - 17$

$y^2 - 2y + 1 = 8x - 16$

$(y-1)^2 = 8(x-2)$

Vertex (2, 1)

ii)



d) $x^3 + 2x^2 + 3x + 1 = 0$

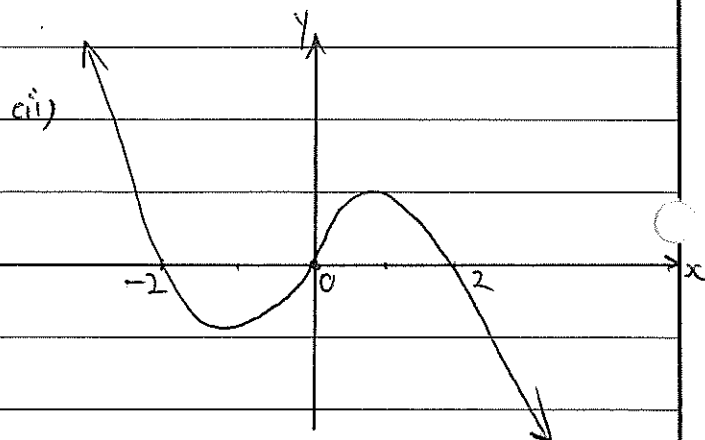
$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$= \left(-\frac{2}{1}\right)^2 - 2 \times \frac{3}{1}$

$= -2$

$$\begin{aligned}
 7. \quad x^2 + 10x - 3 &\equiv ax(x+1) + b(x-1)^2 \\
 &\equiv ax^2 + ax + bx^2 - 2bx + b \\
 &\equiv (a+b)x^2 + (a-2b)x + b \\
 \therefore b &= -3, \quad a+b=1 \quad a-2b=10 \\
 a-3 &= 1 \quad a-2(-3)=10 \\
 \underline{a} &= 4 \quad \underline{a} = 4
 \end{aligned}$$

$$\begin{aligned}
 b) \text{ i) } P(x) &= -x^3 - ax \\
 P(-2) &= -(-2)^3 - a(-2) = 0 \\
 &= 8 + 2a = 0 \\
 \underline{a} &= -4
 \end{aligned}$$



$$\text{ciii) } -2 \leq x \leq 0, \quad x \geq 2$$

$$\begin{aligned}
 d) \text{ i) } 3 \sin x - \cos x &= 1 \\
 3 \times \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} &= 1 \quad \text{where } t = \tan \frac{x}{2} \\
 6t - 1 + t^2 &= 1 + t^2 \\
 6t &= 2 \\
 \underline{t} &= \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{cii) } \tan \frac{x}{2} &= \frac{1}{3} \\
 \frac{x}{2} &= 18^\circ 26' \\
 \underline{x} &= 36^\circ 52'
 \end{aligned}$$

$$\begin{aligned}
 d) \quad \frac{1}{x} - \frac{1}{x-2} &> 0 \\
 x &\neq 0, \quad x \neq 2 \\
 \frac{1}{x} - \frac{1}{x-2} &= 0 \\
 x-2 - x &= 0
 \end{aligned}$$

No solution

$$\underline{0 < x < 2}$$

$$8. a) \sqrt{2} \cos(x+60) = 1$$

$$\cos(x+60) = \frac{1}{\sqrt{2}}$$

$$x+60 = 45^\circ, 315^\circ \text{ and } 05^\circ$$

$$x = 255^\circ, 345^\circ$$

b) Gradients of tangents are 2, -1

$$\tan \theta = \left| \frac{2 - (-1)}{1 + (-2)} \right|$$

$$= \left| \frac{3}{-1} \right|$$

$$= 3$$

$$\theta = 72^\circ$$

$$c) \sec x - \tan x = k$$

$$\sec^2 x - \tan^2 x = k(\sec x + \tan x)$$

$$\text{Now since } \tan^2 x + 1 = \sec^2 x$$

$$1 = k(\sec x + \tan x)$$

$$\therefore \sec x + \tan x = \frac{1}{k}$$

d) If distance (0,0) to $3x - 4y + 10 = 0$ equals 2, line must be a tangent.

$$d = \left| \frac{3 \times 0 + -4 \times 0 + 10}{\sqrt{3^2 + (-4)^2}} \right|$$

$$= \left| \frac{10}{5} \right|$$

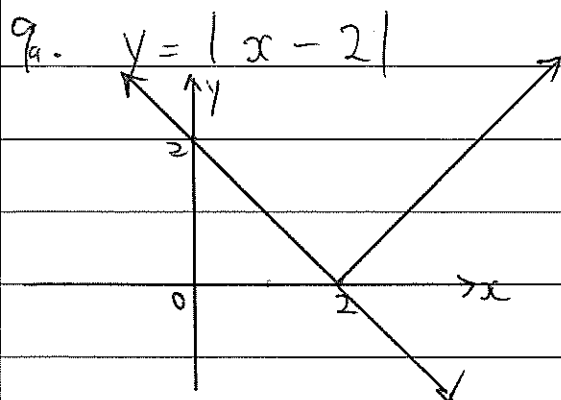
$$d = 2 \quad \checkmark$$

$$e) \text{iv) } \sin(A+B) + \sin(A-B) = 2 \sin A \cos B$$

$$\sin A \cos B + \cancel{\sin B \cos A} + \sin A \cos B - \cancel{\sin B \cos A} = \text{LHS}$$

$$2 \sin A \cos B = \text{LHS} = \text{RHS}$$

$$\begin{aligned}
 \text{cii) } 2 \sin 45 \cos 15 &= \sin(45+15) + \sin(45-15) \\
 &= \sin 60 + \sin 30 \\
 &= \frac{\sqrt{3}}{2} + \frac{1}{2} \\
 &= \frac{1+\sqrt{3}}{2}
 \end{aligned}$$



$$\begin{aligned}
 \text{cii) } |x - 2| &= 2 - x \\
 \text{for } x &\leq 2
 \end{aligned}$$

b) c) $P(2ap, ap^2)$ $Q(2aq, aq^2)$

$$m_{PQ} = \frac{aq^2 - ap^2}{2aq - 2ap}$$

$$= \frac{a(q-p)(q+p)}{2a(q-p)}$$

$$m_{PQ} = \frac{p+q}{2}$$

$$y - ap^2 = \frac{p+q}{2}(x - 2ap)$$

$$2y - 2ap^2 = (p+q)x - 2ap(p+q)$$

$$2y - 2ap^2 = (p+q)x - 2ap^2 - 2apq$$

$$\therefore (p+q)x - 2y = 2apq$$

cii) If $pq = 1$
 $(p+q)x - 2y = 2a$
 when $x = 0$

$$y = -a$$

$$\therefore (0, -a) \text{ fixed}$$

c) $(k+1)x^2 - 2(k-1)x + (k-5)$
 is a perfect square if

$$\Delta = 0$$

$$4(k-1)^2 - 4(k+1)(k-5) = 0$$

$$4k^2 - 8k + 4 - 4(k^2 - 5k + k - 5) = 0$$

$$4k^2 - 8k + 4 - 4k^2 + 20k - 4k + 20 = 0$$

$$8k + 24 = 0$$

$$k = -3$$

$$\begin{aligned}
 \text{d) i) } & \frac{1}{d} + \frac{1}{d^2} + \frac{1}{d^3} \\
 & \frac{\frac{d^2}{d^3} + \frac{d}{d^3} + \frac{1}{d^3}}{d^3} \\
 & = \frac{1 + d + d^2}{d^3} \\
 & = \frac{d^3 + d^3 + d^3}{d^6} \\
 & = \frac{C}{-d}
 \end{aligned}$$

$$\begin{aligned}
 \text{cii) } & d^3 + d^4 + d^5 = C \\
 & d^2(d + d^2 + d^3) = C \\
 & d^2 \times -b = C \\
 & \text{Cube both sides} \\
 & d^6 \times -b^3 = C^3 \\
 & -d \times -b^3 = C^3 \\
 & b^3 d - C^3 = 0
 \end{aligned}$$

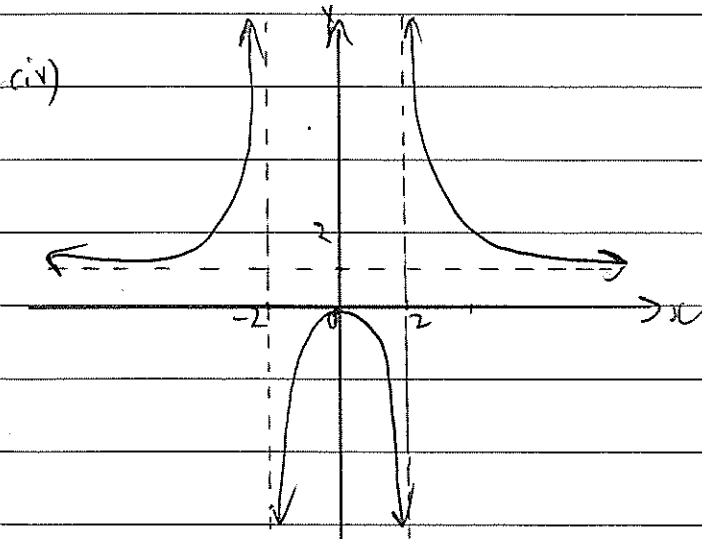
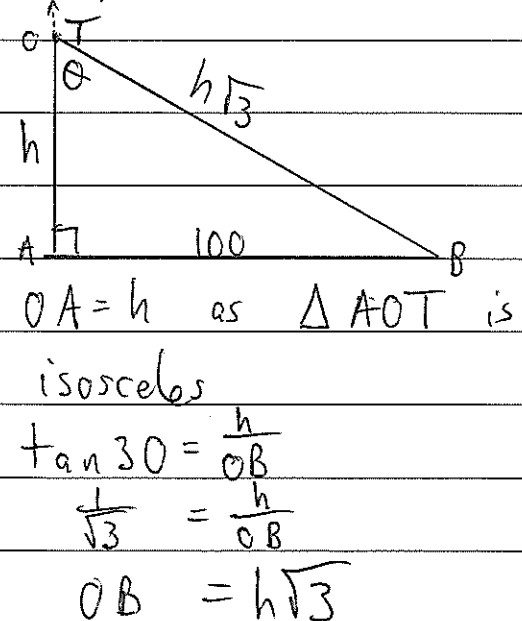
10. g) Not differentiable at $x=1$ because curve doesn't exist at that point.

$$\text{b) i) } \lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 4} = 1$$

$$\begin{aligned}
 \text{cii) } & f(-x) = f(x) \text{ if even} \\
 & \frac{(-x)^2}{(-x)^2 - 4} = \frac{x^2}{x^2 - 4} \\
 & \frac{x^2}{x^2 - 4} = \frac{x^2}{x^2 - 4}
 \end{aligned}$$

ciii) Domain all real x except $x = \pm 2$

c) i) Top View:



$$\text{c.cii) } h^2 + 10000 = 3h^2 \text{ (Pythag.)} \quad \text{cii) } \tan \theta = \frac{100}{50\sqrt{2}} = \sqrt{2}$$

$$h^2 = 5000$$

$$h = 50\sqrt{2}$$

$$\theta = 54^\circ 44' \text{ or}$$

$$\theta = 55^\circ$$

$$11a) f(x) = 2x^3 - 7x^2 - 7x + 30 = (x-3)(2x^2 - x - 10)$$

$$\text{By inspection} = (x-3)(2x-5)(x+2)$$

$$\therefore \text{Solutions are } 3, -2, 2\frac{1}{2}$$

$$b) \text{ Let } \frac{dy}{dx} = 5$$

$$\frac{dy}{dx} = 2x + 3 = 5$$

$$2x = 2$$

$$\therefore \text{ at } x = 1$$

$$(1, 3)$$

$$c) \text{ i) } M \text{ is } \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$$

$$M \left(a(p+q), \frac{a(p^2+q^2)}{2} \right)$$

$$\text{cii) } y = \frac{1}{2}(p+q)x - apq$$

$$\text{Sub. in } (0, a)$$

$$a = 0 - apq$$

$$1 = -pq$$

$$\therefore pq = -1$$

$$\text{ciii) } x^2 = 2ay - 2a^2$$

$$\text{Sub. in } M$$

$$a^2(p+q)^2 = 2a \times \frac{a(p^2+q^2)}{2} - 2a^2$$

$$a^2(p+q)^2 = a^2(p^2+q^2) - 2a^2$$

$$= a^2(p^2+q^2-2)$$

$$= a^2(p^2+q^2+2pq) \text{ from part cii)}$$

$$= a^2(p+q)^2$$

$$\therefore M \text{ lies on this parabola}$$

$$\text{civ) } x^2 = 2ay - 2a^2$$

$$x^2 = 2a(y-a)$$

$$\text{Vertex is } (0, a)$$

$$\text{Focus is } \left(0, \frac{3a}{2} \right)$$